



NITTE
(Deemed to be University)

**NMAM INSTITUTE
OF TECHNOLOGY**

(An off-Campus Institution of NITTE (DEEMED TO BE UNIVERSITY), MANGALORE)

Department of MCA

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ASSIGNMENT 4

1. Write a php script to show the use of various string related functions.

[index.php](#)

```
<?php
$str = "Hello World";

// Length of string
echo "Length: " . strlen($str) . "<br>";

// Word count
echo "Word count: " . str_word_count($str) . "<br>";

// Reverse string
echo "Reverse: " . strrev($str) . "<br>";

// Find position of a word
echo "Position of 'World': " . strpos($str, "World") . "<br>";

// Replace text
echo "Replace: " . str_replace("World", "PHP", $str) . "<br>";

// Convert to uppercase
echo "Uppercase: " . strtoupper($str) . "<br>";

// Convert to lowercase
echo "Lowercase: " . strtolower($str) . "<br>";

// Capitalize first letter
echo "Ucfirst: " . ucfirst("hello world") . "<br>";

// Capitalize each word
echo "Ucwords: " . ucwords("hello world") . "<br>";
?>
```

← → ↻ ⓘ localhost/awt/a4/p1/index.php

Length: 11
Word count: 2
Reverse: dlroW olleH
Position of 'World': 6
Replace: Hello PHP
Uppercase: HELLO WORLD
Lowercase: hello world
Ucfirst: Hello world
Ucwords: Hello World

2. Demonstrate php's support for sorting array elements in php.

[index.php](#)

```
<?php
// Indexed array
$arr = array(5, 2, 9, 1, 7);

// Sort in ascending order
sort($arr);
echo "Ascending order: ";
print_r($arr);
echo "<br>";

// Sort in descending order
rsort($arr);
echo "Descending order: ";
print_r($arr);
echo "<br>";

// Associative array
$marks = array("A" => 85, "B" => 75, "C" => 95);

// Sort by value (ascending)
asort($marks);
echo "Sort by value (asc): ";
print_r($marks);
echo "<br>";

// Sort by key (ascending)
ksort($marks);
echo "Sort by key (asc): ";
print_r($marks);
echo "<br>";
?>
```

← → ↻ ⓘ localhost/awt/a4/p2/index.php

Ascending order: Array ([0] => 1 [1] => 2 [2] => 5 [3] => 7 [4] => 9)

Descending order: Array ([0] => 9 [1] => 7 [2] => 5 [3] => 2 [4] => 1)

Sort by value (asc): Array ([B] => 75 [A] => 85 [C] => 95)

Sort by key (asc): Array ([A] => 85 [B] => 75 [C] => 95)

3. Write a PHP program to count the frequency of each element in an array using functions.

[index.php](#)

```
<?php
```

```
function countFrequency($arr) {
```

```
    $freq = array();
```

```
    foreach ($arr as $value) {
```

```
        if (isset($freq[$value])) {
```

```
            $freq[$value]++;
```

```
        } else {
```

```
            $freq[$value] = 1;
```

```
        }
```

```
    }
```

```
    return $freq;
```

```
}
```

```
// Input array
```

```
$arr = array(1, 2, 2, 3, 3, 3, 4);
```

```
// Call function
```

```
$result = countFrequency($arr);
```

```
// Display result
```

```
echo "Element frequencies:<br>";
```

```
foreach ($result as $key => $value) {
```

```
    echo "$key => $value <br>";
```

```
}
```

```
?>
```

```
← → ↻ ⓘ localhost/awt/a4/p3/index.php

Element frequencies:
1 => 1
2 => 2
3 => 3
4 => 1
```

4. Write a php script to reverse an inputted number and calculate its sum.

[index.php](#)

```
<?php
$num = 12345;

$original = $num;
$reverse = 0;
$sum = 0;

while ($num > 0) {
    $digit = $num % 10; // get last digit
    $reverse = $reverse * 10 + $digit; // build reverse number
    $sum += $digit; // add to sum
    $num = (int)($num / 10); // remove last digit
}

echo "Original Number: $original <br>";
echo "Reversed Number: $reverse <br>";
echo "Sum of digits: $sum <br>";
?>
```

```
← → ↻ ⓘ localhost/awt/a4/p4/index.php

Original Number: 12345
Reversed Number: 54321
Sum of digits: 15
```

5. Input the elements to the array and find second most frequent element in it(without using built in functions).

[index.php](#)

```
<?php
// Input array
```

```

$arr = array(1, 2, 2, 3, 3, 3, 4, 4, 4, 4, 5);

// Step 1: Count frequency manually
$freq = array();

foreach ($arr as $val) {
    if (isset($freq[$val])) {
        $freq[$val]++;
    } else {
        $freq[$val] = 1;
    }
}

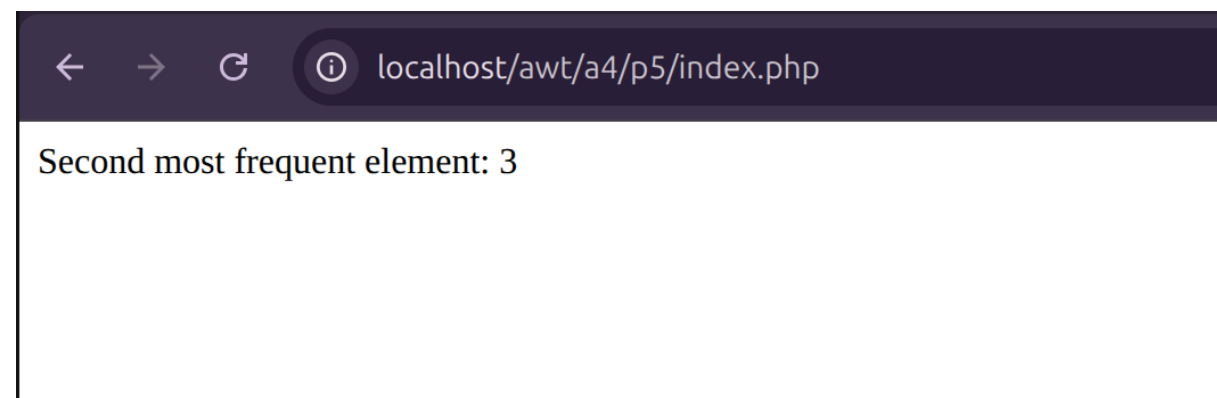
// Step 2: Find highest and second highest frequency
$max = 0;
$secondMax = 0;
$maxElem = null;
$secondElem = null;

foreach ($freq as $key => $count) {
    if ($count > $max) {
        $secondMax = $max;
        $secondElem = $maxElem;

        $max = $count;
        $maxElem = $key;
    } elseif ($count > $secondMax && $count < $max) {
        $secondMax = $count;
        $secondElem = $key;
    }
}

// Output
echo "Second most frequent element: $secondElem";
?>

```



6. Write a PHP program to find the maximum and the minimum in array (without using built in functions).

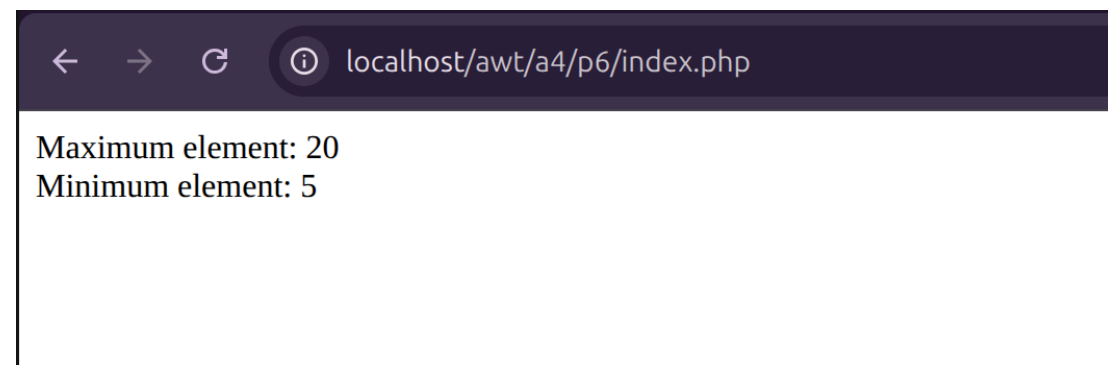
[index.php](#)

```
<?php
// Input array
$arr = array(10, 5, 20, 8, 15);

// Assume first element as max and min
$max = $arr[0];
$min = $arr[0];

// Traverse array
foreach ($arr as $value) {
    if ($value > $max) {
        $max = $value;
    }
    if ($value < $min) {
        $min = $value;
    }
}

// Output
echo "Maximum element: $max <br>";
echo "Minimum element: $min <br>";
?>
```



7. Write a PHP program to accept name, email, and age from an HTML form and display the submitted data and also validate the data using pattern matching

[form.html](#)

```
<!DOCTYPE html>
<html>
<head>
    <title>User Form</title>
</head>
<body>
<form method="post" action="process.php">
    Name: <input type="text" name="name"><br><br>
```

```
Email: <input type="text" name="email"><br><br>
Age: <input type="text" name="age"><br><br>
<input type="submit" name="submit" value="Submit">
</form>
</body>
</html>
```

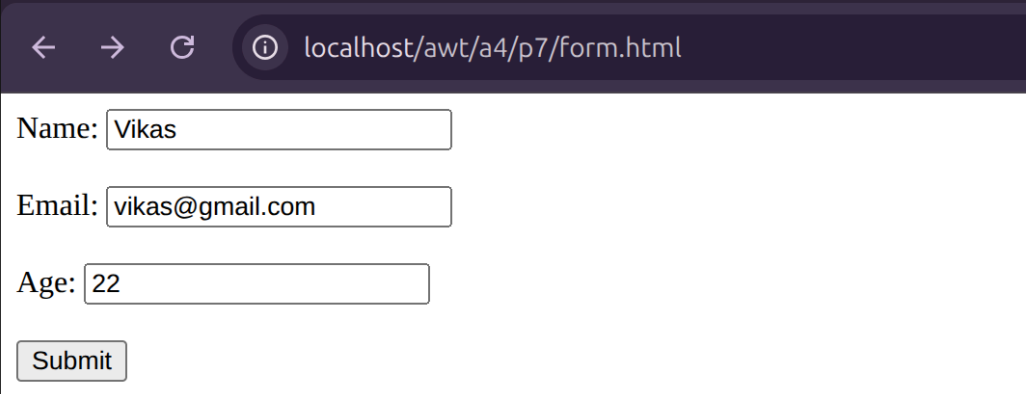
[process.php](#)

```
<?php
if (isset($_POST['submit'])) {

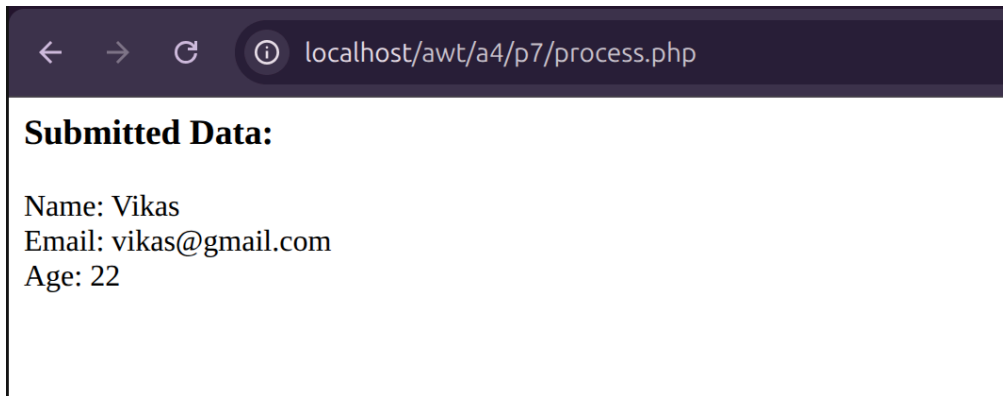
    $name = $_POST['name'];
    $email = $_POST['email'];
    $age = $_POST['age'];

    // Regex patterns
    $namePattern = "/^[a-zA-Z ]+$/";
    $emailPattern = "/^[\\w\\.]+@[\\w\\.]+\\. [a-zA-Z]{2,}$/";
    $agePattern = "/^[0-9]{1,3}$/";

    // Validation
    if (!preg_match($namePattern, $name)) {
        echo "Invalid Name<br>";
    } elseif (!preg_match($emailPattern, $email)) {
        echo "Invalid Email<br>";
    } elseif (!preg_match($agePattern, $age)) {
        echo "Invalid Age<br>";
    } else {
        echo "<h3>Submitted Data:</h3>";
        echo "Name: $name <br>";
        echo "Email: $email <br>";
        echo "Age: $age <br>";
    }
}
?>
```



The screenshot shows a web browser window with the address bar displaying 'localhost/awt/a4/p7/form.html'. The page content includes a form with three input fields: 'Name:' with the value 'Vikas', 'Email:' with the value 'vikas@gmail.com', and 'Age:' with the value '22'. Below these fields is a 'Submit' button.



8. Write a PHP program to count the number of words in a file.

[index.php](#)

```
<?php
$filename = "sample.txt";

// Check if file exists
if (file_exists($filename)) {

    // Read file content
    $content = file_get_contents($filename);

    // Count words manually
    $wordCount = 0;
    $words = preg_split("/\s+/", trim($content));

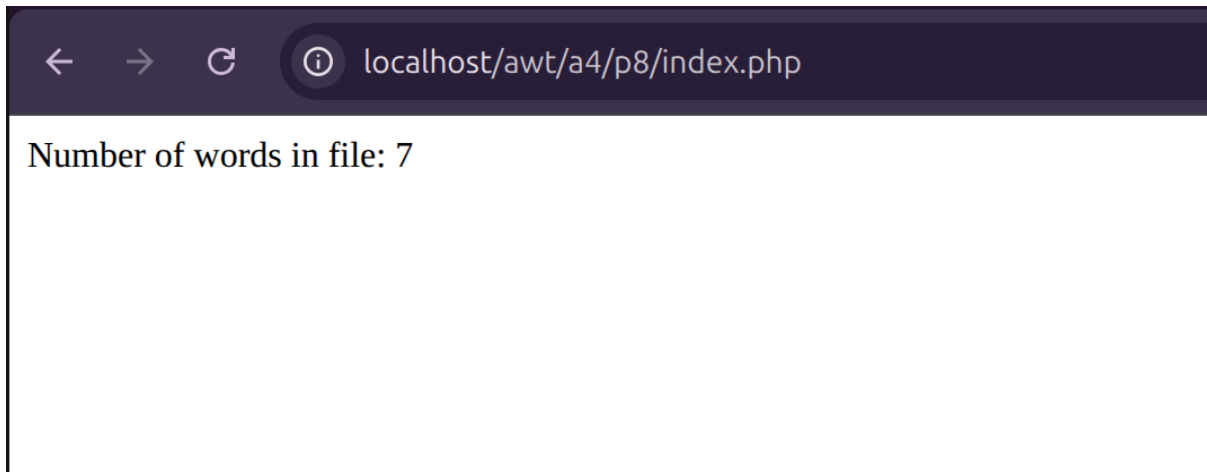
    foreach ($words as $word) {
        if ($word != "") {
            $wordCount++;
        }
    }

    echo "Number of words in file: $wordCount";
} else {
    echo "File does not exist.";
}
?>
```

[sample.txt](#)

Hello world

This is a PHP program



9. Write a PHP program to create and retrieve a cookie storing the username.

[set_cookie.php](#)

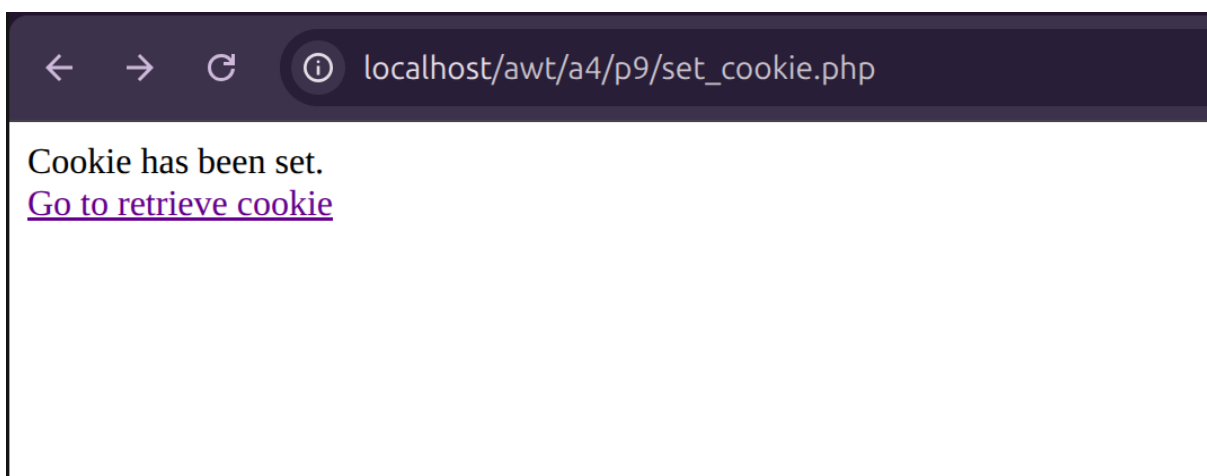
```
<?php
$username = "Shashi";

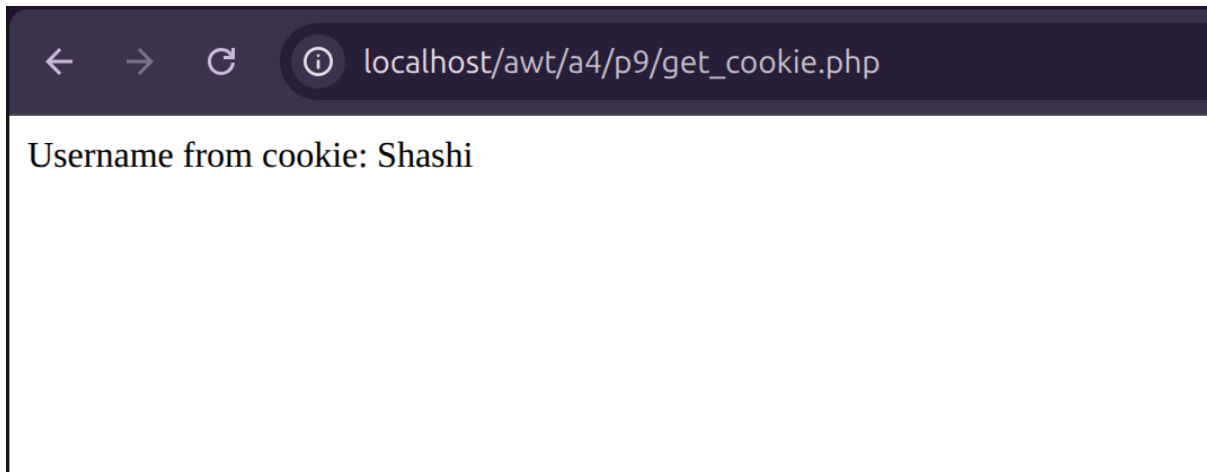
// Set cookie (expires in 1 hour)
setcookie("user", $username, time() + 3600);

echo "Cookie has been set. <br>";
echo "<a href='get_cookie.php'>Go to retrieve cookie</a>";
?>
```

[get_cookie.php](#)

```
<?php
if (isset($_COOKIE['user'])) {
    echo "Username from cookie: " . $_COOKIE['user'];
} else {
    echo "Cookie not set!";
}
?>
```





10. Write a PHP program to demonstrate session tracking by storing and displaying user login information.

login.php

```
<?php
session_start();

if (isset($_POST['submit'])) {
    $_SESSION['username'] = $_POST['username'];
    echo "Login successful! <br>";
    echo "<a href='dashboard.php'>Go to Dashboard</a>";
}
?>

<form method="post">
    Username: <input type="text" name="username">
    <input type="submit" name="submit" value="Login">
</form>
```

dashboard.php

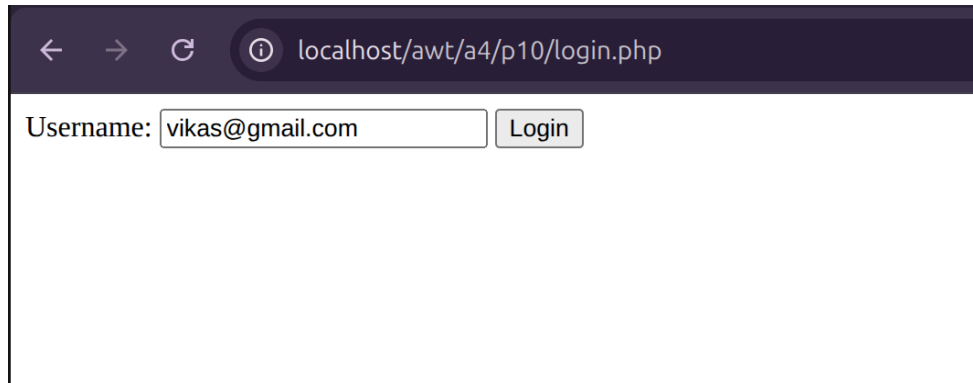
```
<?php
session_start();

if (isset($_SESSION['username'])) {
    echo "Welcome, " . $_SESSION['username'] . "<br>";
    echo "<a href='logout.php'>Logout</a>";
} else {
    echo "Please login first.";
}
?>
```

logout.php

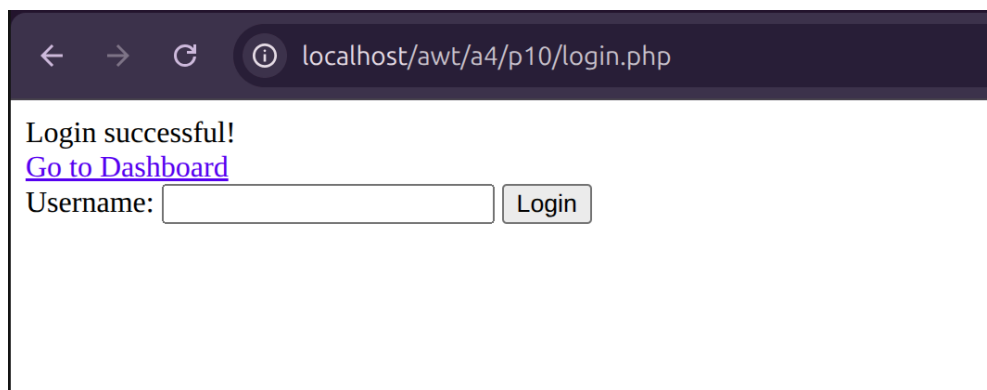
```
<?php
session_start();
session_destroy();

echo "You have been logged out.<br>";
echo "<a href='login.php'>Login again</a>";
?>
```



localhost/awt/a4/p10/login.php

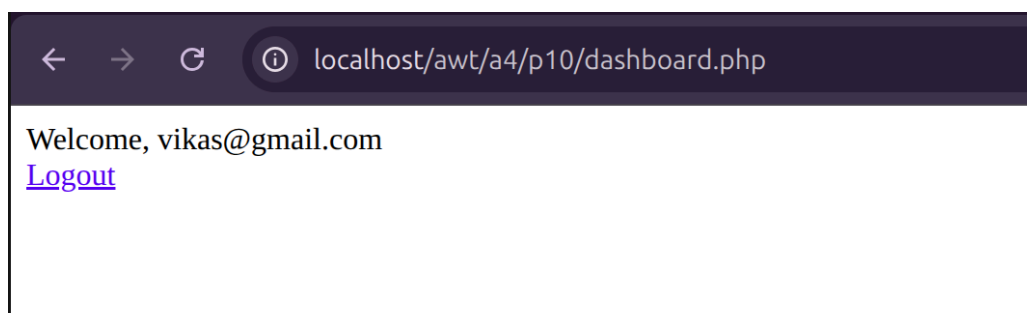
Username:



localhost/awt/a4/p10/login.php

Login successful!
[Go to Dashboard](#)

Username:



localhost/awt/a4/p10/dashboard.php

Welcome, vikas@gmail.com
[Logout](#)

11. Write a PHP program to create a table in a MySQL database using PHP and show the CRUD operations on it

MySQL Queries (Run in XAMPP phpMyAdmin)

-- Create Database

```
CREATE DATABASE testdb;
```

-- Use Database

```
USE testdb;
```

```
-- Create Table
CREATE TABLE users (
    id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(50),
    email VARCHAR(50)
);
```

[crud.php](#)

```
<?php
// DB Connection
$conn = new mysqli("localhost", "root", "", "testdb");

if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}

// INSERT
if (isset($_POST['insert'])) {
    $name = $_POST['name'];
    $email = $_POST['email'];

    $conn->query("INSERT INTO users (name, email) VALUES ('$name', '$email')");
}

// UPDATE
if (isset($_POST['update'])) {
    $id = $_POST['id'];
    $name = $_POST['name'];

    $conn->query("UPDATE users SET name='$name' WHERE id=$id");
}

// DELETE
if (isset($_POST['delete'])) {
    $id = $_POST['id'];

    $conn->query("DELETE FROM users WHERE id=$id");
}
?>

<!DOCTYPE html>
<html>
<body>

<h3>CRUD Operations</h3>

<!-- FORM -->
<form method="post">
```

```

ID (for update/delete): <input type="text" name="id"><br><br>
Name: <input type="text" name="name"><br><br>
Email: <input type="text" name="email"><br><br>

<input type="submit" name="insert" value="Insert">
<input type="submit" name="update" value="Update">
<input type="submit" name="delete" value="Delete">
</form>

<hr>

<h3>All Records</h3>

<?php
// READ
$result = $conn->query("SELECT * FROM users");

while($row = $result->fetch_assoc()) {
    echo $row['id'] . " - " . $row['name'] . " - " . $row['email'] . "<br>";
}
?>

</body>
</html>

```

Insert

←
→
↺
🔍 localhost/awt/a4/p11/crud.php

CRUD Operations

ID (for update/delete):

Name:

Email:

All Records

CRUD Operations

ID (for update/delete):

Name:

Email:

All Records

1 - Vikas - vikas@gmail.com

2 - Prakash - prakash@gmail.com

Update

CRUD Operations

ID (for update/delete):

Name:

Email:

All Records

1 - Vikas - vikas@gmail.com

2 - Prakash - prakash@gmail.com

All Records

1 - Vikas SR - vikas@gmail.com

2 - Prakash - prakash@gmail.com

[Delete](#)

CRUD Operations

ID (for update/delete):

Name:

Email:

All Records

2 - Prakash - prakash@gmail.com